

CHEM-1310
Fall 2017
Second Examination
Form A

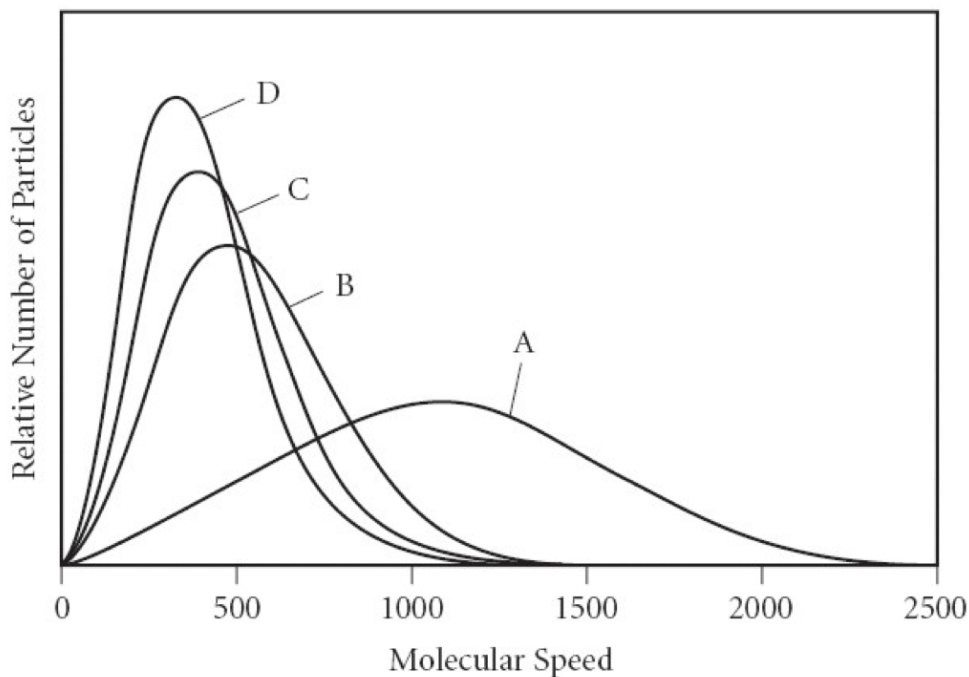
Multiple Choice - Circle the BEST Answer

1. Which of the following molecules in the gas phase will have the slowest rate of effusion?
 - A) F_2
 - B) Br_2
 - C) CO_2
 - D) CH_4
 - E) All molecules have the same rate of effusion.

2. Which of the following statement(s) about the kinetic-molecular theory is/are true?
 - A) The kinetic-molecular theory describes that when molecules collide their collisions are elastic.
 - B) The kinetic-molecular theory describes how gas molecules are so tiny that they do not occupy a volume of their own.
 - C) The kinetic-molecular theory states that the temperature in Kelvin of the gas is dependent on the average kinetic energy of that gas.
 - D) Only A and B are true
 - E) A, B and C are all true statements.

Multiple Choice - Circle the BEST Answer

3. The following graph represents four different gases at the same temperature. Which of the following statements are true?



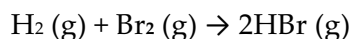
- A) Molecules of the gas represented by curve A have a greater distribution of speeds due to the gas having a larger molar mass.
- B) Molecules of the gas represented by curve C have a smaller molar mass than the molecules represent by gas B.
- C) All of the curves represented are actually the same molecule and at the same temperature, but have different root mean square speeds.
- D) Molecules of the gas represented by curve D have the smallest root mean square speed and would diffuse the fastest.
- E) None of the above statements are true.
4. The molar heat capacity of a compound with the formula C_2H_6SO is $88.0 \text{ J/mol}\cdot\text{K}$. The specific heat of this substance is _____ $\text{J/g}\cdot\text{K}$.
- A) 6.88×10^3
- B) 1.13
- C) 4.89
- D) 88.0
- E) -88.0

Multiple Choice - Circle the BEST Answer

5. Which of the following conditions will always result in an increase in the internal energy of the system?

- A) system loses heat and does work on the surroundings
- B) system gains heat and does work on the surroundings
- C) system loses heat and has work done on it by the surroundings
- D) system gains heat and has work done on it by the surroundings
- E) none of the above

6. The value of ΔH° for the reaction below is -72 kJ. How many kJ of heat are released when 1.0 mol of HBr is formed in this reaction.



- A) -144
- B) 144
- C) 72
- D) 0.44
- E) 36

7. The value of ΔH° for the following reaction is 177.8 kJ. The value of ΔH_f° for CaO(s) is _____ kJ/mol.

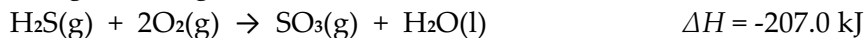


Substance	ΔH_f° (kJ/mol)
CO (g)	-110.5
CO ₂ (g)	-393.7
CaCO ₃ (s)	-1207.0

- A) -1600
- B) -635.5
- C) 813.4
- D) 177.8
- E) -813.4

Multiple Choice - Circle the BEST Answer

8. Given the following reactions



What is the enthalpy of the following reaction?



- A) - 388.5 kJ
- B) -109
- C) +28.5 kJ
- D) +72.5 kJ
- E) -15.5

9. 1.150 g of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ goes through combustion in a bomb calorimeter. If the temperature increased from 23.42°C to 27.64°C and the heat capacity of the calorimeter is $4.90 \text{ kJ}/^\circ\text{C}$ what is enthalpy of combustion in kJ/mol of sucrose?

- A) -20.68 kJ/mol
- B) $-6.152 \times 10^3 \text{ kJ/mol}$
- C) 23.78 kJ/mol
- D) 17.98 kJ/mol
- E) 6100 kJ/mol

**Covered on
Exam 1
202008**

10. What is the wavelength of light in meters that has a frequency of $1.20 \times 10^{13} \text{ s}^{-1}$?

- A) 25.0 m
- B) 2.5 m
- C) $2.50 \times 10^{-5} \text{ m}$
- D) 0.0400 m
- E) 12.0 m

Multiple Choice - Circle the BEST Answer

11. What is the wavelength of a photon with energy of $3.7 \times 10^{-18} \text{ J}$?

- A) $5.4 \times 10^{-8} \text{ m}$
- B) $2.2 \times 10^3 \text{ m}$
- C) $1.9 \times 10^7 \text{ m}$
- D) $1.9 \times 10^{-25} \text{ m}$
- E) $3.7 \times 10^{-21} \text{ m}$

12. What is the wavelength (in m) of an electron whose velocity is $1.7 \times 10^4 \text{ m/s}$ and whose mass is $9.1 \times 10^{-31} \text{ kg}$?

- A) 12 m
- B) $4.3 \times 10^{-11} \text{ m}$
- C) $4.3 \times 10^{-8} \text{ m}$
- D) $2.3 \times 10^{-7} \text{ m}$
- E) $2.3 \times 10^7 \text{ m}$

13. Of the following transitions in the Bohr hydrogen atom, which transition results in the emission of the highest energy photon?

- A) $n=3 \rightarrow n=5$
- B) $n=5 \rightarrow n=1$
- C) $n=5 \rightarrow n=3$
- D) $n=1 \rightarrow n=5$
- E) $n=1 \rightarrow n=4$

14. What are the quantum numbers for the outermost electron in a strontium atom (Sr) in the ground state assuming convention? (n, l, m_l, m_s)

- A) 4, 3, 3, $+1/2$
- B) 3, 1, -1, $-1/2$
- C) 3, 0, 0, $+1/2$
- D) 5, 0, 0, $-1/2$
- E) 5, 1, 0, $+1/2$

Multiple Choice - Circle the BEST Answer

15. How many orbitals are present in the third energy level of an atom?

- A) 2
- B) 3
- C) 4
- D) 9
- E) 10

16. How many p-orbitals are occupied in the ground state of silicon (Si)?

- A) 3
- B) 4
- C) 5
- D) 6
- E) 7

17. In a ground state iron atom the _____ subshell is partially filled.

- A) 3s
- B) 3p
- C) 3d
- D) 4s
- E) 4p

18. In which orbital does an electron in an atom experience the greatest nuclear charge?

- A) 3d
- B) 3p
- C) 3s
- D) 2p
- E) 2s

Multiple Choice - Circle the BEST Answer

19. Which of the following correctly represents the first ionization energy of bromine?

- A) $\text{Br (g)} \rightarrow \text{Br}^+(\text{g}) + \text{e}^-$
- B) $\text{Br}_2 (\text{g}) + 2\text{e}^- \rightarrow 2 \text{Br}^- (\text{g})$
- C) $\text{Br (g)} + \text{e}^- \rightarrow \text{Br}^- (\text{g})$
- D) $\text{Br}^+ (\text{g}) + \text{e}^- \rightarrow \text{Br (g)}$
- E) $\text{Br}_2 (\text{g}) + \text{e}^- \rightarrow \text{Br}^- (\text{g})$

20. Of the following species which has the largest radius?

- A) Rb^+
- B) Sr^{2+}
- C) Br^-
- D) Kr
- E) Ar

21. Which form of the exam do you have?

- A) A
- B) B